

CSE400: Fundamentals of Probability in Computing

Lecture 20: Linear Transformations and Expectations of Multiple RVs

March 19, 2025

1 Outline

- Statistical Representation of Random Vectors
- Mean Vector, Correlation, and Covariance Matrices
- Linear Transformations
- Mapping properties under $Y = AX + b$
- Multivariate Gaussian Distributions
- From 1D/2D cases to N-dimensional Joint PDFs
- Special Cases and Properties
- Implications of zero correlation and matrix factorization

2 Statistical Representation of Random Vectors

2.1 Definition

Let $\mathbf{X}$ be an N -dimensional random vector:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

3 Mean Vector

3.1 Definition

The mean vector is:

$$\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}] = \begin{bmatrix} \mathbb{E}[X_1] \\ \mathbb{E}[X_2] \\ \vdots \\ \mathbb{E}[X_N] \end{bmatrix}$$

4 Covariance Matrix

4.1 Definition

The covariance matrix is:

$$\Sigma = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T]$$

4.2 Interpretation

- Represents dependencies between variables
- Acts as a “dependency map”

5 Multivariate Gaussian Distribution

5.1 Assumption

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

5.2 Implication

The system is defined by:

- Mean (typical conditions)
- Covariance (interdependencies)

6 Linear Transformations

6.1 Definition

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$$

6.2 Mapping Properties

- Linear transformation of random vector

7 Case Study: Smart City Traffic + Weather System

7.1 Step 1: Define Random Variables

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{bmatrix}$$

(N-dimensional random vector, here $N = 5$)

7.2 Step 2: Mean Vector

$$\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$$

7.3 Step 3: Covariance Matrix Interpretation

- More rain $\Rightarrow$ more congestion
- More rain $\Rightarrow$ lower speed
- More traffic $\Rightarrow$ higher pollution
- Higher temperature $\Rightarrow$ pollution disperses

7.4 Step 4: Joint Gaussian Distribution

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

7.5 Step 6: Transformation

Define congestion index:

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

7.6 Inference

If rainfall increases:

- Speed ↓
- Traffic ↑
- Pollution ↑